Quiz 12: Soil, Impressions & Advanced Prints

Review of Session 12. ~15 minutes. A soil texture triangle helps.

Section A: Soil

1. Give the particle-size range for sand, silt, and clay.
2. Using the texture triangle, name the class:
   - 60% sand, 30% silt, 10% clay
   - 20% sand, 60% silt, 20% clay
   - 40% sand, 40% silt, 20% clay
   - 25% sand, 25% silt, 50% clay
3. What causes soil to be red? Dark brown/black? Gray/blue-gray? White/pale?
4. Is soil class evidence or individual evidence? What can a soil comparison do?

Section B: Fingerprints

1. List the eight IAFIS fingerprint patterns.
2. How many deltas does each have: arch, loop, whorl?
3. Approximate frequency of loops, whorls, arches in the population?
4. What is the difference between a latent, a patent, and a plastic print?
5. Name two ways to develop a latent print.
6. Define: dermatoglyphics, dactyloscopy, adermatoglyphia, hexadactyly.
7. In which layer of skin do fingerprint ridges form?
8. What does CODIS stand for? Is it a fingerprint database or a DNA database?

Section C: Footwear

1. Give one class characteristic and one individual characteristic of a shoeprint.
2. Same size and same tread design in two prints — does that prove they're the same shoe?

Answer Key

Section A

1. Sand 2.0–0.05 mm; silt 0.05–0.002 mm; clay < 0.002 mm.
2. Sandy loam; silt loam; loam; clay.
3. Red = iron oxides; dark brown/black = organic matter; gray/blue = waterlogged (reduced iron); white/pale = carbonates, salts, or lime.
4. Class evidence — a comparison of texture, color, pH, and mineral content can place a suspect at a scene or rule them out, but rarely proves one exact location.

Section B

1. Plain arch, tented arch, radial loop, ulnar loop, plain whorl, central pocket loop whorl, double loop whorl, accidental whorl.
2. Arch 0, loop 1, whorl 2.
3. Loops ~60–65%, whorls ~30–35%, arches ~5%.
4. Latent = invisible (sweat/oil, must be developed); patent = visible (finger was coated in blood/ink/grease and transferred a print); plastic = 3-D impression in a soft material.
5. Dusting powder; superglue (cyanoacrylate) fuming; ninhydrin; iodine fuming; fluorescent dye (any two).
6. Dermatoglyphics = the study of friction-ridge skin patterns; dactyloscopy = fingerprint identification; adermatoglyphia = born with no fingerprints; hexadactyly = born with a sixth finger or toe.
7. The dermal papillae — the papillary layer of the dermis.
8. Combined DNA Index System; a DNA database.

Section C

1. Class: size, tread design, brand. Individual: a specific cut, gouge, worn spot, or embedded pebble.
2. No — that's class evidence; you need a matching individual characteristic to identify the specific shoe.